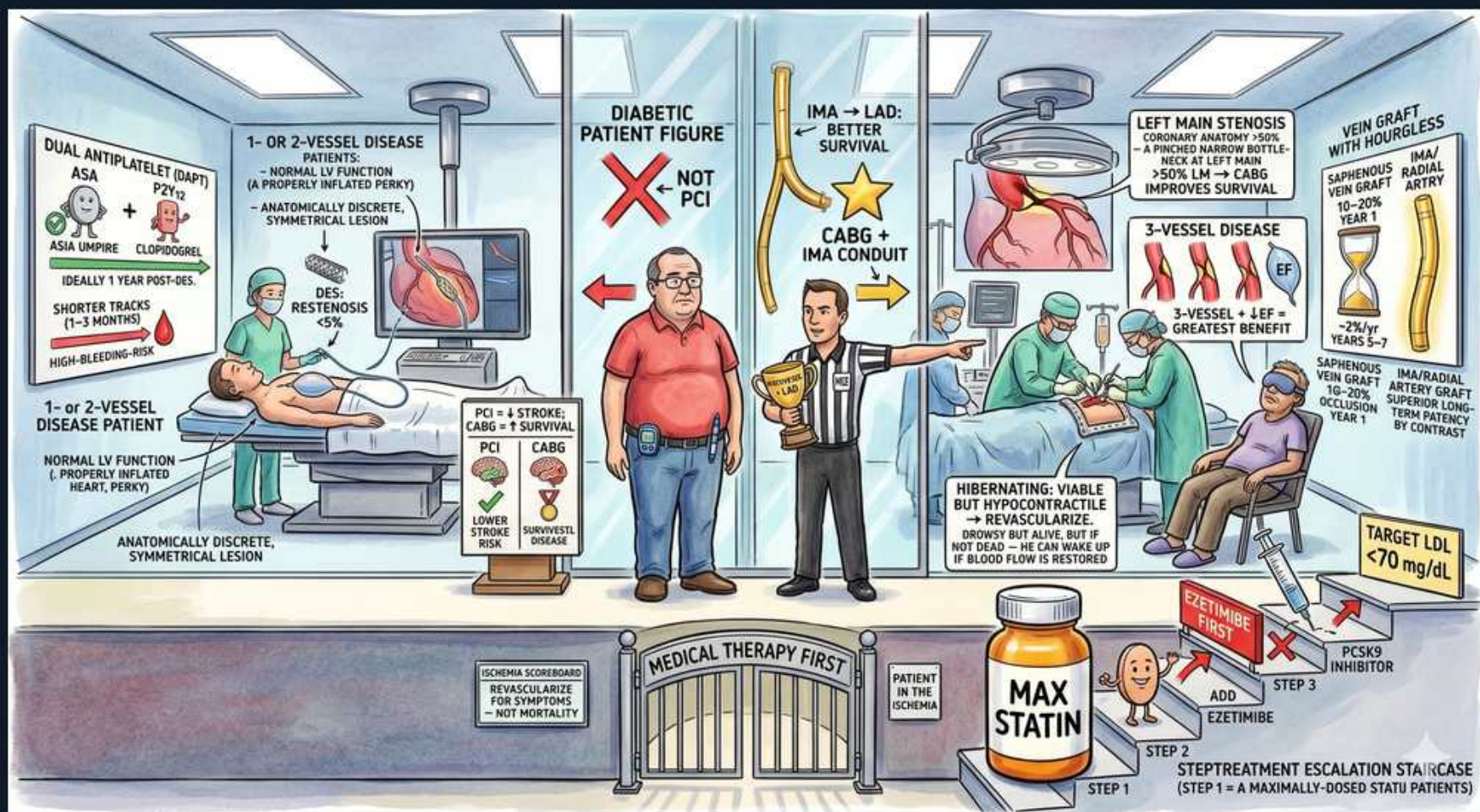


IHD — Revascularization: PCI vs CABG & Medical Therapy



1. Medical therapy first

WHAT YOU SEE

Bottom banner 'MEDICAL THERAPY FIRST', vault door

WHAT IT ENCODES

Optimal medical therapy (OMT) is the foundation for ALL chronic IHD — high-intensity statin, aspirin, antianginals (beta-blocker/CCB/nitrate/ranolazine), and risk-factor control (BP, glucose, smoking). Revascularization is added on top of OMT for persistent symptoms despite therapy or for specific high-risk anatomy (left main, 3-vessel + low EF, diabetic multivessel). COURAGE and ISCHEMIA showed that for stable CAD, an initial invasive strategy did not reduce death/MI vs OMT. Mnemonic: 'medicate before you dilate.'

BOARD TRAP

WRONG answer: rushing a stable-angina patient with normal EF straight to catheterization/PCI before a trial of optimal medical therapy. Discriminator — ISCHEMIA/COURAGE proved upfront PCI has no mortality/MI advantage over OMT in stable CAD; revascularization is for refractory symptoms or high-risk anatomy.

2. Revasc relieves ischemia, not mortality

WHAT YOU SEE

Small sign 'REVASCULARIZE → RELIEVE THE ISCHEMIA, NOT MORTALITY'

WHAT IT ENCODES

In STABLE CAD, revascularization (especially PCI) relieves angina and ischemia but does NOT reduce mortality or MI compared with optimal medical therapy — proven by COURAGE (2007) and ISCHEMIA (2020). The exceptions where revascularization DOES improve survival are left main, 3-vessel disease with reduced EF, and diabetic multivessel disease (and of course ACS/STEMI, where emergent reperfusion saves lives). Mnemonic: 'in stable disease, stents soothe symptoms, not survival.'

BOARD TRAP

WRONG answer: telling a stable-angina patient that PCI will help them 'live longer.' Discriminator — PCI in stable IHD improves quality of life/angina only; the survival-benefit settings are CABG in high-risk anatomy and emergent reperfusion in ACS. ISCHEMIA confirmed no mortality benefit of routine invasive strategy.

3. 1–2 vessel disease → PCI

WHAT YOU SEE

Left 'PCI: 1- OR 2-VESSEL DISEASE — anatomically discrete, symmetrical lesion'

WHAT IT ENCODES

PCI is best suited to 1–2 vessel disease with discrete, focal, symmetrical lesions, preserved LV function, and a low SYNTAX score — these are technically straightforward to stent with good durability. PCI is less invasive, has lower periprocedural stroke risk than CABG, and allows rapid recovery. Long, calcified, ostial, bifurcation, or chronic-total-occlusion lesions (high SYNTAX) favor CABG instead. Mnemonic: 'few vessels + focal lesions = percutaneous.'

BOARD TRAP

WRONG answer: applying PCI to complex, diffuse, or high-SYNTAX multivessel anatomy just because it is non-surgical. Discriminator — SYNTAX score quantifies complexity; high scores predict PCI failure/restenosis and favor CABG. PCI's sweet spot is discrete, low-complexity 1–2 vessel disease.

4. Dual antiplatelet after PCI

WHAT YOU SEE

'DUAL ANTIPLATELET (DAPT): ASA + P2Y12 (clopidogrel)', 'ideally 1 year post-DES'

WHAT IT ENCODES

After PCI with a drug-eluting stent, dual antiplatelet therapy = aspirin 81 mg daily (lifelong) PLUS a P2Y12 inhibitor (clopidogrel 75 mg, or ticagrelor 90 mg BID / prasugrel in ACS). Standard DAPT duration is ~6 months for stable CAD and ~12 months after ACS, to prevent stent thrombosis while the stent endothelializes. Premature DAPT cessation is the leading cause of catastrophic stent thrombosis. Mnemonic: 'Aspirin Always + P2Y12 for a year.'

BOARD TRAP

WRONG answer: stopping the P2Y12 inhibitor early (e.g., for elective non-cardiac surgery) within the mandatory DAPT window. Discriminator — early DAPT discontinuation after DES causes acute stent thrombosis (often a fatal STEMI); elective surgery should be deferred past the DAPT period or aspirin continued perioperatively.

5. DES restenosis vs shorter DAPT

WHAT YOU SEE

'DES = RESTENOSIS', 'shorter tracks (1–3 months)'

WHAT IT ENCODES

Drug-eluting stents (DES) elute antiproliferative drugs (e.g., everolimus, sirolimus) that suppress neointimal hyperplasia, dramatically reducing in-stent restenosis vs bare-metal stents (BMS). Newer-generation DES also re-endothelialize faster, permitting SHORTER DAPT (as little as 1–3 months in high-bleeding-risk patients). Stent type and bleeding risk together determine DAPT duration. Mnemonic: 'DES = Drug stops rE-Stenosis.'

BOARD TRAP

WRONG answer: assuming bare-metal stents are 'safer' because they need shorter DAPT. Discriminator — modern DES have both lower restenosis AND now allow short DAPT; BMS are largely obsolete. The reason to ever shorten DAPT is high bleeding risk, not stent superiority.

6. Diabetic + multivessel → NOT PCI

WHAT YOU SEE

'DIABETIC PATIENT FIGURE — NOT PCI' with red X

WHAT IT ENCODES

Diabetics with multivessel CAD do better with CABG than PCI — the FREEDOM trial showed CABG reduced death/MI vs DES-PCI (NNT ~12 at 5 yr). Diabetic coronary disease is diffuse, distal, and progresses fast with high in-stent restenosis, so segmental stenting fails to keep pace; arterial grafts bypass the whole diseased territory. Mnemonic: 'Diabetic Diffuse Disease → Detour with CABG.'

BOARD TRAP

WRONG answer: choosing multivessel PCI for a diabetic to avoid surgery. Discriminator — FREEDOM is the landmark proof that CABG is superior for hard outcomes in diabetics with multivessel disease; PCI is reserved for single-vessel/focal disease or prohibitive surgical risk.

7. PCI: ↓stroke but ↓survival

WHAT YOU SEE

'PCI = ↓STROKE, ↓SURVIVAL vs CABG', 'lower stroke risk'

WHAT IT ENCODES

Compared with CABG, PCI has a LOWER periprocedural stroke risk (no aortic cannulation/cross-clamp, no cardiopulmonary bypass) but LOWER long-term survival in complex/multivessel disease and a higher need for repeat revascularization. This is the central trade-off the boards test: CABG buys durability/survival at the cost of upfront stroke and surgical morbidity. Mnemonic: 'PCI dodges the stroke today but loses the war over years.'

BOARD TRAP

WRONG answer: citing PCI's lower stroke risk to justify it over CABG in high-risk anatomy. Discriminator — the lower stroke risk is real but is outweighed by CABG's survival benefit and durability in left main/3-vessel/diabetic disease; stroke risk only tips the scale in borderline or high-surgical-risk cases.

8. CABG + IMA → better survival

WHAT YOU SEE

'LAD → BETTER SURVIVAL', 'CABG + IMA CONDUIT', trophy + star

WHAT IT ENCODES

CABG using the internal mammary artery grafted to the LAD (IMA→LAD) is the cornerstone of surgical revascularization: arterial conduits resist atherosclerosis (>90% patency at 10 yr) and provide an independent survival benefit in left main and multivessel disease. The IMA stays attached to the subclavian (in-situ graft), adapting flow to demand. Mnemonic: 'IMA is immortal — the artery that grants survival.'

BOARD TRAP

WRONG answer: treating saphenous vein grafts as equivalent to the IMA. Discriminator — the survival advantage of CABG is driven by the arterial IMA→LAD graft; vein grafts fail ~50% by 10 yr. A CABG without an IMA graft loses much of its long-term edge.

9. 3-vessel disease → CABG

WHAT YOU SEE

'3-VESSEL DISEASE', surgeons operating, '3-VESSEL + LOW EF = GREATEST BENEFIT'

WHAT IT ENCODES

Three-vessel disease (and left main) favors CABG, with the GREATEST survival benefit when EF is reduced (STICH: CABG improved long-term survival in ischemic cardiomyopathy with EF <35%). CABG completely revascularizes all territories with durable grafts, whereas multivessel PCI leaves higher restenosis and incomplete revascularization. Higher SYNTAX score → stronger case for surgery. Mnemonic: '3 vessels + weak pump = surgeon's strongest indication.'

BOARD TRAP

WRONG answer: medical-therapy-alone or multivessel PCI for 3-vessel disease with EF <35%. Discriminator — STICH extended follow-up favored CABG for survival in ischemic cardiomyopathy; this is one of the few settings where revascularization (surgical) truly prolongs life.

10. Vein graft hourglass (fails sooner)

WHAT YOU SEE

'VEIN GRAFT WITH HOURGLASS — saphenous vein vs radial artery'

WHAT IT ENCODES

Saphenous vein grafts occlude far sooner than arterial grafts: ~10–20% in year 1, ~2%/yr thereafter, accelerating after year 7 (~50% patent at 10 yr) due to endothelial injury, intimal hyperplasia, and vein-graft atherosclerosis. Arterial conduits (IMA, radial) have superior patency, so surgeons maximize arterial grafting. Aggressive statin + aspirin therapy slows vein-graft disease. Mnemonic: 'the hourglass runs out on vein grafts — arteries endure.'

BOARD TRAP

WRONG answer: attributing recurrent angina years after CABG to a 'new problem' rather than graft failure. Discriminator — late post-CABG angina classically reflects saphenous vein-graft attrition/atherosclerosis; the IMA graft is usually still patent. Arterial grafts are the durable solution.

11. Hibernating myocardium = viable

WHAT YOU SEE

'HIBERNATING: VIABLE BUT HYPOCONTRACTILE → REVASCULARIZE', '↓EF but recovers'

WHAT IT ENCODES

Hibernating myocardium is chronically underperfused but VIABLE tissue that has down-regulated contraction to survive — it appears akinetic/hypocontractile (low EF) but RECOVERS function after revascularization. Viability is confirmed by dobutamine stress echo (contractile reserve), PET, or cardiac MRI (delayed enhancement <50% wall thickness). Identifying it justifies revascularizing a low-EF heart. Mnemonic: 'hibernating = sleeping, not dead — wake it with blood flow.'

BOARD TRAP

WRONG answer: dismissing akinetic, low-EF segments as scar/dead myocardium and withholding revascularization. Discriminator — viable hibernating tissue (positive viability study) recovers contractility after CABG/PCI; only non-viable transmural scar (delayed enhancement >50% on MRI) will not benefit.

12. Max statin (step 1)

WHAT YOU SEE

'MAX STATIN', 'STEP 1', staircase

WHAT IT ENCODES

Lipid-lowering step 1 for established ASCVD is a maximally-tolerated HIGH-intensity statin — atorvastatin 40–80 mg or rosuvastatin 20–40 mg — aiming for ≥50% LDL reduction. Statins inhibit HMG-CoA reductase and have pleiotropic plaque-stabilizing effects. Every CAD patient needs one unless truly intolerant. Mnemonic: 'statin is step one and forever.'

BOARD TRAP

WRONG answer: starting a low/moderate-intensity statin (or no statin) in a patient with established CAD. Discriminator — secondary prevention mandates high-intensity statin for ≥50% LDL lowering and target LDL <70 (often <55); under-dosing the statin is a common board error.

13. Add ezetimibe (step 2)

WHAT YOU SEE

'ADD EZETIMIBE — STEP 2'

WHAT IT ENCODES

If LDL remains above goal (<70, often <55 mg/dL) on maximally-tolerated high-intensity statin, step 2 is to ADD ezetimibe 10 mg daily — it blocks intestinal cholesterol absorption (NPC1L1 transporter) and lowers LDL a further ~15–25%. IMPROVE-IT showed adding ezetimibe to statin reduced cardiovascular events post-ACS. Mnemonic: 'Ezetimibe Eats up gut cholesterol — step 2.'

BOARD TRAP

WRONG answer: jumping to an expensive PCSK9 inhibitor before maximizing statin and trying ezetimibe. Discriminator — guideline-directed escalation is statin → ezetimibe → PCSK9 inhibitor; ezetimibe is the cheaper, oral step 2 with proven event reduction (IMPROVE-IT).

14. PCSK9 inhibitor (step 3) / LDL <70

WHAT YOU SEE

'PCSK9 INHIBITOR — STEP 3', 'TARGET LDL <70 mg/dL'

WHAT IT ENCODES

Step 3 for persistently high LDL despite max statin + ezetimibe is a PCSK9 inhibitor (evolocumab or alirocumab, subcutaneous q2–4 wk) — these monoclonal antibodies block PCSK9 so more LDL receptors recycle to the hepatocyte surface, lowering LDL ~50–60% further. FOURIER/ODYSSEY showed event reduction in high-risk ASCVD. Secondary-prevention LDL target is <70 mg/dL, and <55 in very-high-risk patients. Mnemonic: 'PCSK9 inhibitor = more LDL receptors recycled, LDL plummets.'

BOARD TRAP

WRONG answer: aiming only for LDL <100 in a patient with established CAD, or skipping further therapy once <100 is reached. Discriminator — established ASCVD targets LDL <70 (often <55 in very-high-risk); PCSK9 inhibitors exist precisely to reach these aggressive goals when statin + ezetimibe fall short.